



Problem Solving With C
UE24CS151B

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1. **(1 Mark)** What header file is required to use the `printf()` and `scanf()` functions in C?
2. **(1 Mark)** What is the default output stream used by the `printf()` function?
3. **(2 Marks)** Explain the difference between ordinary characters and conversion specifications in the format string of a `printf()` statement.
4. **(2 Marks)** Why is the address operator (`&`) necessary when using `scanf()` to read values into variables of primary types?
5. **(3 Marks)** Describe what escape sequences are and provide three examples.
6. **(3 Marks)** Explain what the `sizeof` operator does and its relevance to understanding data types in C.
7. **(4 Marks)** What are the possible return values of the `scanf()` function, and what does each value indicate about the input process?
8. **(5 Marks)** Explain the purpose of format specifiers (like `%d`, `%f`, `%c`, and `%s`) in `printf()` and `scanf()`. Give an example of how each is used and what type of data they correspond to.
9. **(5 Marks)** Describe the components of a `printf()` format string, including flags, field width, precision, and conversion character. Provide an example of how to use each of these components.
10. **(10 Marks)** Write a C program snippet that does the following:
 - Prompts the user to enter their name (a string).
 - Prompts the user to enter their age (an integer).
 - Prints the entered information to the console in a formatted way, including an appropriate message.